Direct Sums and Orthogonal Complements

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Sums and direct sums

> **Source**
>
> This section, Section and Section are adapted from Zhou [1], used under the MIT License (see the license text in Section in Subspaces and Rank). The source proves that a subspace and its orthogonal complement together make up $\mathbb{R}^p$ by extending an orthonormal basis; the proof here uses the rank-nullity theorem (Theorem 8 in Column Space, Null Space and Rank-Nullity) instead. The source defines the orthogonal complement inside an ambient vector space $V \subseteq \mathbb{R}^m$; here the ambient space is always $\mathbb{R}^p$, which is all that the fundamental theorem and the projections need.

Definition 0.1 (Sum of subspaces)**.** The **sum** of two subspaces $\mathcal{S}_1$ and $\mathcal{S}_2$ of $\mathbb{R}^p$ (Definition 4 in Subspaces and Rank) is the set of all sums of a vector from $\mathcal{S}_1$ and a vector from $\mathcal{S}_2$:

$$\mathcal{S}_1 + \mathcal{S}_2 \stackrel{\text{def}}{=} \{\underset{\sim}{x}_1 + \underset{\sim}{x}_2 : \underset{\sim}{x}_1 \in \mathcal{S}_1,\ \underset{\sim}{x}_2 \in \mathcal{S}_2\}.$$

Example 0.1 (Two lines sum to a plane)**.** Let $\mathcal{S}_1 = \text{span}\{(1,0,0)\}$ and $\mathcal{S}_2 = \text{span}\{(0,1,0)\}$, two lines in $\mathbb{R}^3$. A vector of $\mathcal{S}_1 + \mathcal{S}_2$ has the form $a\,(1,0,0) + b\,(0,1,0) = (a,b,0)$, and a and b can be any numbers, so $\mathcal{S}_1 + \mathcal{S}_2 = \{(a,b,0) : a,b \in \mathbb{R}\}$, the plane $z = 0$.

Example 0.2 (The sum is not the union)**.** For the two lines in Example 0.1, the union $\mathcal{S}_1 \cup \mathcal{S}_2$ contains $(1,0,0)$ and $(0,1,0)$ but not their sum $(1,1,0)$: every multiple of $(1,0,0)$ has second entry 0, and every multiple of $(0,1,0)$ has first entry 0. So the union is not closed under addition and is not a subspace, while the sum, the whole plane $z = 0$, is.

Exercise 0.1 (Testing whether a vector is in a sum of two lines)**.** A vector is in the sum of two subspaces when it splits into one piece from each. Let $\mathcal{S}_1 = \text{span}\{(1,0,1)\}$ and $\mathcal{S}_2 = \text{span}\{(0,1,1)\}$, two lines in $\mathbb{R}^3$.
Decide whether the vector $(2,3,5)$ is in $\mathcal{S}_1 + \mathcal{S}_2$.

Hint

Look for a and b with $a\,(1,0,1) + b\,(0,1,1) = (2,3,5)$, starting with the first two entries.

Solution

Solution 0.1. Here $a\,(1,0,1) + b\,(0,1,1) = (a, b, a+b)$, whose first two entries give $a = 2$ and $b = 3$, and then the third entry is $2+3 = 5$. So $(2,3,5) = 2\,(1,0,1) + 3\,(0,1,1)$, with $2\,(1,0,1) \in \mathcal{S}_1$ and $3\,(0,1,1) \in \mathcal{S}_2$, and $(2,3,5)$ is in $\mathcal{S}_1 + \mathcal{S}_2$.

Theorem 0.1 (Sums and intersections of subspaces are subspaces)**.** *If $\mathcal{S}_1$ and $\mathcal{S}_2$ are subspaces of $\mathbb{R}^p$ (Definition 4 in Subspaces and Rank), then $\mathcal{S}_1 + \mathcal{S}_2$ (Definition 0.1) and $\mathcal{S}_1 \cap \mathcal{S}_2$ are subspaces of $\mathbb{R}^p$.*

Proof

Proof. Both $\mathcal{S}_1$ and $\mathcal{S}_2$ contain $\underset{\sim}{0}$ (Theorem 1 in Subspaces and Rank).
The sum. It is not empty: it contains $\underset{\sim}{0} + \underset{\sim}{0} = \underset{\sim}{0}$. Take two of its vectors, $\underset{\sim}{u} = \underset{\sim}{u}_1 + \underset{\sim}{u}_2$ and $\underset{\sim}{w} = \underset{\sim}{w}_1 + \underset{\sim}{w}_2$, with $\underset{\sim}{u}_1, \underset{\sim}{w}_1 \in \mathcal{S}_1$ and $\underset{\sim}{u}_2, \underset{\sim}{w}_2 \in \mathcal{S}_2$, and a number c. Then

$$\begin{aligned} \underset{\sim}{u} + \underset{\sim}{w} &= (\underset{\sim}{u}_1 + \underset{\sim}{u}_2) + (\underset{\sim}{w}_1 + \underset{\sim}{w}_2) && \text{(substitute)} \\ &= (\underset{\sim}{u}_1 + \underset{\sim}{w}_1) + (\underset{\sim}{u}_2 + \underset{\sim}{w}_2), && \text{(regroup; vector addition is entrywise)} \end{aligned}$$

where $\underset{\sim}{u}_1 + \underset{\sim}{w}_1 \in \mathcal{S}_1$ and $\underset{\sim}{u}_2 + \underset{\sim}{w}_2 \in \mathcal{S}_2$, because each subspace is closed under addition. Likewise

$$\begin{aligned} c\,\underset{\sim}{u} &= c\,(\underset{\sim}{u}_1 + \underset{\sim}{u}_2) && \text{(substitute)} \\ &= c\,\underset{\sim}{u}_1 + c\,\underset{\sim}{u}_2, && \text{(distribute } c \text{ entrywise)} \end{aligned}$$

where $c\,\underset{\sim}{u}_1 \in \mathcal{S}_1$ and $c\,\underset{\sim}{u}_2 \in \mathcal{S}_2$, because each subspace is closed under scalar multiplication. So $\underset{\sim}{u} + \underset{\sim}{w}$ and $c\,\underset{\sim}{u}$ are in $\mathcal{S}_1 + \mathcal{S}_2$.
The intersection. It is not empty: it contains $\underset{\sim}{0}$. If $\underset{\sim}{u}$ and $\underset{\sim}{w}$ are in both $\mathcal{S}_1$ and $\mathcal{S}_2$, then $\underset{\sim}{u} + \underset{\sim}{w}$ and $c\,\underset{\sim}{u}$ are in $\mathcal{S}_1$, because $\mathcal{S}_1$ is a subspace, and in $\mathcal{S}_2$ for the same reason, so they are in $\mathcal{S}_1 \cap \mathcal{S}_2$. □

Example 0.3 (Two planes in $\mathbb{R}^3$)**.** Let $\mathcal{S}_1 = \{(a, b, 0) : a, b \in \mathbb{R}\}$, the plane $z = 0$, and $\mathcal{S}_2 = \{(0, b, c) : b, c \in \mathbb{R}\}$, the plane $x = 0$.

- **Intersection:** a vector in both has third entry 0 and first entry 0, so $\mathcal{S}_1 \cap \mathcal{S}_2 = \{(0, b, 0) : b \in \mathbb{R}\}$, the y-axis, which is a line through the origin.
- **Sum:** $(a, b, 0)+(0, b', c) = (a, b+b', c)$, and any (x, y, z) arises this way, with $a = x$, $b = y$, $b' = 0$ and $c = z$. So $\mathcal{S}_1 + \mathcal{S}_2 = \mathbb{R}^3$.

Exercise 0.2 (Finding the intersection of two planes)**.** Theorem 0.1 says that two subspaces meet in a subspace. Let $\mathcal{S}_1 = \{(a, b, 0) : a, b \in \mathbb{R}\}$, the plane $z = 0$ in $\mathbb{R}^3$, and $\mathcal{S}_2 = \text{span}\{(1, 1, 0), (0, 0, 1)\}$.
Find $\mathcal{S}_1 \cap \mathcal{S}_2$.

Hint

Express a vector of $\mathcal{S}_2$ as $s\,(1, 1, 0) + t\,(0, 0, 1)$, and ask when its third entry is 0.

Solution

Solution 0.2. A vector of $\mathcal{S}_2$ is $s\,(1, 1, 0) + t\,(0, 0, 1) = (s, s, t)$. It is in $\mathcal{S}_1$ exactly when its third entry is 0, that is, when $t = 0$. So

$$\mathcal{S}_1 \cap \mathcal{S}_2 = \{(s, s, 0) : s \in \mathbb{R}\} = \text{span}\{(1, 1, 0)\},$$

a line through the origin, which is a subspace, as Theorem 0.1 says.

Theorem 0.2 (The dimension of a sum of subspaces)**.** *If $\mathcal{S}_1$ and $\mathcal{S}_2$ are subspaces of $\mathbb{R}^p$, then*

$$\dim(\mathcal{S}_1 + \mathcal{S}_2) = \dim(\mathcal{S}_1) + \dim(\mathcal{S}_2) - \dim(\mathcal{S}_1 \cap \mathcal{S}_2).$$

Proof

Proof. All three sets in the formula are subspaces (Theorem 0.1), so each has a dimension (Definition 9 in Subspaces and Rank). Let $k = \dim(\mathcal{S}_1 \cap \mathcal{S}_2)$, $d_1 = \dim(\mathcal{S}_1)$ and $d_2 = \dim(\mathcal{S}_2)$.
Build a list. Choose a basis $\underset{\sim}{z}_1, \dots, \underset{\sim}{z}_k$ of $\mathcal{S}_1 \cap \mathcal{S}_2$ (Theorem 9 in Subspaces and Rank). These vectors are linearly independent and lie in $\mathcal{S}_1$, so Theorem 9 in Subspaces and Rank extends them to a basis $\underset{\sim}{z}_1, \dots, \underset{\sim}{z}_k, \underset{\sim}{x}_1, \dots, \underset{\sim}{x}_{d_1-k}$ of $\mathcal{S}_1$, which has d_1 vectors (Theorem 7 in Subspaces and Rank). In the same way, extend them to a basis $\underset{\sim}{z}_1, \dots, \underset{\sim}{z}_k, \underset{\sim}{y}_1, \dots, \underset{\sim}{y}_{d_2-k}$

of $\mathcal{S}_2$. We show that the $\underset{\sim}{z}$'s, $\underset{\sim}{x}$'s and $\underset{\sim}{y}$'s together, $k + (d_1 - k) + (d_2 - k) = d_1 + d_2 - k$ vectors, are a basis of $\mathcal{S}_1 + \mathcal{S}_2$ (Definition 6 in Subspaces and Rank).

They span $\mathcal{S}_1 + \mathcal{S}_2$. Any $\underset{\sim}{v} \in \mathcal{S}_1 + \mathcal{S}_2$ is $\underset{\sim}{v}_1 + \underset{\sim}{v}_2$ with $\underset{\sim}{v}_1 \in \mathcal{S}_1$ and $\underset{\sim}{v}_2 \in \mathcal{S}_2$. $\underset{\sim}{v}_1$ is a linear combination of the $\underset{\sim}{z}$'s and $\underset{\sim}{x}$'s, and $\underset{\sim}{v}_2$ is a linear combination of the $\underset{\sim}{z}$'s and $\underset{\sim}{y}$'s, so their sum is a linear combination of all three groups. Conversely, each vector in the list is in $\mathcal{S}_1 + \mathcal{S}_2$ (for example, $\underset{\sim}{x}_1 = \underset{\sim}{x}_1 + \underset{\sim}{0}$), and $\mathcal{S}_1 + \mathcal{S}_2$ is a subspace, so it contains every linear combination of them.

They are linearly independent. Suppose

$$\sum_{i=1}^{k} a_i \underset{\sim}{z}_i + \sum_{j=1}^{d_1-k} b_j \underset{\sim}{x}_j + \sum_{l=1}^{d_2-k} c_l \underset{\sim}{y}_l = \underset{\sim}{0}.$$

Let $\underset{\sim}{w} \stackrel{\text{def}}{=} \sum_l c_l \underset{\sim}{y}_l$. Then $\underset{\sim}{w} \in \mathcal{S}_2$, because it is a linear combination of vectors of $\mathcal{S}_2$. Subtracting $\sum_i a_i \underset{\sim}{z}_i + \sum_j b_j \underset{\sim}{x}_j$ from both sides of the supposed equation gives

$$\underset{\sim}{w} = \sum_{i=1}^{k} (-a_i)\, \underset{\sim}{z}_i + \sum_{j=1}^{d_1-k} (-b_j)\, \underset{\sim}{x}_j,$$

a linear combination of vectors of $\mathcal{S}_1$, so $\underset{\sim}{w} \in \mathcal{S}_1$ too. So $\underset{\sim}{w} \in \mathcal{S}_1 \cap \mathcal{S}_2$, and $\underset{\sim}{w} = \sum_i e_i \underset{\sim}{z}_i$ for some numbers e_i, since the $\underset{\sim}{z}$'s span the intersection. Then

$$\begin{aligned}
\underset{\sim}{0} &= \underset{\sim}{w} - \underset{\sim}{w} && \text{(a vector minus itself)} \\
&= \sum_{i=1}^{k} e_i \underset{\sim}{z}_i - \sum_{l=1}^{d_2-k} c_l \underset{\sim}{y}_l && \text{(substitute the two expressions for } \underset{\sim}{w}\text{)} \\
&= \sum_{i=1}^{k} e_i \underset{\sim}{z}_i + \sum_{l=1}^{d_2-k} (-c_l)\, \underset{\sim}{y}_l, && \text{(write each subtraction as adding } -1 \text{ times)}
\end{aligned}$$

and because the $\underset{\sim}{z}$'s and $\underset{\sim}{y}$'s are a basis of $\mathcal{S}_2$, every $c_l = 0$. The supposed equation then reads $\sum_i a_i \underset{\sim}{z}_i + \sum_j b_j \underset{\sim}{x}_j \stackrel{\sim}{=} \underset{\sim}{0}$, and because the $\underset{\sim}{z}$'s and $\underset{\sim}{x}$'s are a basis of $\mathcal{S}_1$, every a_i and b_j is 0.

So $\dim(\mathcal{S}_1 + \mathcal{S}_2) = d_1 + d_2 - k$. When some of the groups are empty (for example $k = 0$, or $\mathcal{S}_1 = \mathcal{S}_1 \cap \mathcal{S}_2$), the sums over them are $\underset{\sim}{0}$ and the argument goes through unchanged. □

Example 0.4 (Two planes in $\mathbb{R}^3$, by dimension)**.** For the planes $z = 0$ and $x = 0$ in Example 0.3, each has dimension 2: the plane $z = 0$ has basis $(1, 0, 0), (0, 1, 0)$ (Example 16 in Subspaces and Rank), and the plane $x = 0$ has basis $(0, 1, 0), (0, 0, 1)$ by the same argument. Their intersection, the y-axis span$\{(0, 1, 0)\}$, has dimension 1, because the single nonzero vector $(0, 1, 0)$ is linearly independent ($c\,(0, 1, 0) = (0, c, 0)$ is $\underset{\sim}{0}$ only if $c = 0$) and so is a basis of it. Theorem 0.2 gives

$$\begin{aligned}\dim(\mathcal{S}_1 + \mathcal{S}_2) &= 2 + 2 - 1 \\ &= 3,\end{aligned}$$

which agrees with $\mathcal{S}_1 + \mathcal{S}_2 = \mathbb{R}^3$ (Example 0.3) and $\dim(\mathbb{R}^3) = 3$ (Example 16 in Subspaces and Rank).

Corollary 0.1 (The dimension of a sum is at most the sum of the dimensions)**.** *If $\mathcal{S}_1$ and $\mathcal{S}_2$ are subspaces of $\mathbb{R}^p$, then*

$$\dim(\mathcal{S}_1 + \mathcal{S}_2) \leq \dim(\mathcal{S}_1) + \dim(\mathcal{S}_2),$$

with equality exactly when $\mathcal{S}_1 \cap \mathcal{S}_2 = \{\underset{\sim}{0}\}$.

Proof

Proof. By Theorem 0.2, the two sides differ by $\dim(\mathcal{S}_1 \cap \mathcal{S}_2) \geq 0$. That dimension is 0 exactly when a basis of the intersection is the empty list, that is, when the intersection is the span of the empty list, $\{\underset{\sim}{0}\}$ (Definition 5 in Subspaces and Rank, Example 16 in Subspaces and Rank). □

Example 0.5 (Equality and strict inequality)**.**

- For the lines $\mathcal{S}_1 = \text{span}\{(1,0,0)\}$ and $\mathcal{S}_2 = \text{span}\{(0,1,0)\}$ in Example 0.1, a common vector satisfies $a\,(1,0,0) = b\,(0,1,0)$, so $(a,-b,0) = \underset{\sim}{0}$ and

  $$\begin{aligned}a &= b \\ &= 0.\end{aligned}$$

  The intersection is $\{\underset{\sim}{0}\}$, and indeed

  $$\begin{aligned}\dim(\mathcal{S}_1 + \mathcal{S}_2) &= 2 \\ &= 1 + 1 :\end{aligned}$$

  the sum is the plane $z = 0$ (Example 0.1), which has dimension 2 (Example 16 in Subspaces and Rank), and each line has dimension 1, with its spanning vector as a basis.

- For

  $$\begin{aligned}\mathcal{S}_1 &= \mathcal{S}_2 \\ &= \text{span}\{(1,0,0)\},\end{aligned}$$

the sum is the same line, because $a\,(1,0,0)+b\,(1,0,0)=(a+b)\,(1,0,0)$, so $\dim(\mathcal{S}_1+\mathcal{S}_2)=1<1+1$, a strict inequality.

Exercise 0.3 (Finding the dimension of a sum of two planes in $\mathbb{R}^4$)**.** Theorem 0.2 finds the dimension of a sum of two subspaces without describing it. Let $\mathcal{S}_1=\operatorname{span}\{\underset{\sim}{e}_1,\underset{\sim}{e}_2\}$ and $\mathcal{S}_2=\operatorname{span}\{\underset{\sim}{e}_2,\underset{\sim}{e}_3\}$, two planes in $\mathbb{R}^4$, where $\underset{\sim}{e}_1,\dots,\underset{\sim}{e}_4$ are the indicator vectors of $\mathbb{R}^4$. Find $\dim(\mathcal{S}_1+\mathcal{S}_2)$ from Theorem 0.2.

Hint

First find $\mathcal{S}_1\cap\mathcal{S}_2$ by comparing entries.

Solution

Solution 0.3. A vector of $\mathcal{S}_1$ is $(a,b,0,0)$, and a vector of $\mathcal{S}_2$ is $(0,c,d,0)$. They coincide exactly when $a=0$, $d=0$ and $b=c$, so $\mathcal{S}_1\cap\mathcal{S}_2=\operatorname{span}\{\underset{\sim}{e}_2\}$, of dimension 1. Each plane has dimension 2, since two distinct indicator vectors are linearly independent. By Theorem 0.2,

$$\dim(\mathcal{S}_1+\mathcal{S}_2)=2+2-1=3.$$

Exercise 0.4 (Finding the dimension of a sum of subspaces that meet only at the origin)**.** Corollary 0.1 says when the dimension of a sum matches the total of the dimensions. Let $\mathcal{S}_1$ and $\mathcal{S}_2$ be subspaces of $\mathbb{R}^5$ with $\dim(\mathcal{S}_1)=2$, $\dim(\mathcal{S}_2)=3$ and $\mathcal{S}_1\cap\mathcal{S}_2=\{\underset{\sim}{0}\}$, so that they meet only at the origin.
Find $\dim(\mathcal{S}_1+\mathcal{S}_2)$.

Hint

Use the equality case of Corollary 0.1.

Solution

Solution 0.4. Here $\mathcal{S}_1\cap\mathcal{S}_2=\{\underset{\sim}{0}\}$, so Corollary 0.1 holds with equality:

$$\dim(\mathcal{S}_1+\mathcal{S}_2)=2+3=5.$$

Since $\mathcal{S}_1+\mathcal{S}_2\subseteq\mathbb{R}^5$ and both have dimension 5, the two are equal: $\mathcal{S}_1+\mathcal{S}_2=\mathbb{R}^5$.

Theorem 0.3 (Rank is subadditive)**.** *For $m\times n$ matrices* $\mathbf{A}$ *and* $\mathbf{B}$*,*

$$\operatorname{rank}(\mathbf{A}+\mathbf{B}) \leq \operatorname{rank}(\mathbf{A}) + \operatorname{rank}(\mathbf{B}).$$

Proof

Proof. $\mathcal{C}(\mathbf{A}+\mathbf{B}) \subseteq \mathcal{C}(\mathbf{A}) + \mathcal{C}(\mathbf{B})$**.** Any vector of $\mathcal{C}(\mathbf{A}+\mathbf{B})$ is $(\mathbf{A}+\mathbf{B})\underset{\sim}{x}$ for some $\underset{\sim}{x} \in \mathbb{R}^n$ (Definition 1 in Column Space, Null Space and Rank-Nullity), and $(\mathbf{A}+\mathbf{B})\underset{\sim}{x} = \mathbf{A}\underset{\sim}{x} + \mathbf{B}\underset{\sim}{x}$ (Theorem 8 in Matrices, with $\underset{\sim}{x}$ as an $n \times 1$ matrix), a vector of $\mathcal{C}(\mathbf{A})$ plus a vector of $\mathcal{C}(\mathbf{B})$.

Compare dimensions. All the sets involved are subspaces of $\mathbb{R}^m$ (Theorem 1 in Column Space, Null Space and Rank-Nullity, Theorem 0.1), so

$$\begin{aligned}
\operatorname{rank}(\mathbf{A}+\mathbf{B}) &= \dim(\mathcal{C}(\mathbf{A}+\mathbf{B})) && \text{(Theorem 7 in Column Space, Null Space and Rank-Nullity)} \\
&\leq \dim(\mathcal{C}(\mathbf{A}) + \mathcal{C}(\mathbf{B})) && \text{(the containment, and part 2 of Theorem 10 in Subspaces and Ra} \\
&\leq \dim(\mathcal{C}(\mathbf{A})) + \dim(\mathcal{C}(\mathbf{B})) && \text{(Corollary 0.1)} \\
&= \operatorname{rank}(\mathbf{A}) + \operatorname{rank}(\mathbf{B}). && \text{(Theorem 7 in Column Space, Null Space and Rank-Nullity)}
\end{aligned}$$

□

Example 0.6 (Subadditivity with equality and without)**.** Let $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$. Each has one nonzero column and one zero column. The nonzero column on its own is linearly independent ($c\,\underset{\sim}{v} = \underset{\sim}{0}_2$ with $\underset{\sim}{v} \neq \underset{\sim}{0}_2$ forces $c = 0$), and the two columns together are not, since 1 times the zero column is $\underset{\sim}{0}_2$; so each matrix has rank 1 (Definition 2 in Subspaces and Rank). By the same argument, $-\mathbf{A}$ has rank 1.

- $\mathbf{A}+\mathbf{B}$ is the 2×2 identity matrix, which has rank 2 (Example 13 in Column Space, Null Space and Rank-Nullity), so the bound $2 \leq 1+1$ holds with equality.
- $\mathbf{A}+(-\mathbf{A})$ is the 2×2 zero matrix, which has rank 0: any nonempty list of its columns contains $\underset{\sim}{0}_2$, and $1 \cdot \underset{\sim}{0}_2 = \underset{\sim}{0}_2$. So the bound $0 \leq \operatorname{rank}(\mathbf{A}) + \operatorname{rank}(-\mathbf{A}) = 1+1$ is strict.

Exercise 0.5 (Checking subadditivity of rank on two rank-one matrices)**.** Theorem 0.3 says that $\operatorname{rank}(\mathbf{A}+\mathbf{B}) \leq \operatorname{rank}(\mathbf{A}) + \operatorname{rank}(\mathbf{B})$ for matrices of the same size. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 0 & 1 \\ 0 & 3 \end{bmatrix}.$$

Find $\operatorname{rank}(\mathbf{A}+\mathbf{B})$.

Hint

Each of $\mathbf{A}$ and $\mathbf{B}$ has rank 1, so Theorem 0.3 allows at most 2.

Solution

Solution 0.5. In $\mathbf{A}$, the second column is twice the first, so $\text{rank}(\mathbf{A}) = 1$. In $\mathbf{B}$, the first column is $\underset{\sim}{0}_2$ and the second is not, so $\text{rank}(\mathbf{B}) = 1$. Then

$$\mathbf{A} + \mathbf{B} = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}.$$

Its columns are linearly independent: $c_1(1,2) + c_2(3,7) = \underset{\sim}{0}_2$ gives $c_1 = -3c_2$ from the first entry, and then $-6c_2 + 7c_2 = c_2 = 0$ from the second. So $\text{rank}(\mathbf{A} + \mathbf{B}) = 2 = 1 + 1$, and Theorem 0.3 holds with equality.

Definition 0.2 (Complementary subspaces)**.** Let $\mathcal{S}_1$, $\mathcal{S}_2$ and $\mathcal{V}$ be subspaces of $\mathbb{R}^p$. $\mathcal{S}_1$ and $\mathcal{S}_2$ are **complementary** in $\mathcal{V}$ if

$$\mathcal{V} = \mathcal{S}_1 + \mathcal{S}_2 \quad \text{and} \quad \mathcal{S}_1 \cap \mathcal{S}_2 = \{\underset{\sim}{0}\}$$

(Definition 0.1).

Definition 0.3 (Direct sum)**.** If $\mathcal{S}_1$ and $\mathcal{S}_2$ are complementary in $\mathcal{V}$ (Definition 0.2), then $\mathcal{V}$ is the **direct sum** [2] of $\mathcal{S}_1$ and $\mathcal{S}_2$, written $\mathcal{V} = \mathcal{S}_1 \oplus \mathcal{S}_2$.

Example 0.7 ($\mathbb{R}^3$ is a plane plus a line)**.** Let $\mathcal{S}_1 = \{(a, b, 0) : a, b \in \mathbb{R}\}$, the plane $z = 0$, and $\mathcal{S}_2 = \{(0, 0, c) : c \in \mathbb{R}\}$, the z-axis.

- **Sum:** any $(x, y, z) \in \mathbb{R}^3$ is $(x, y, 0) + (0, 0, z)$, so $\mathcal{S}_1 + \mathcal{S}_2 = \mathbb{R}^3$.
- **Intersection:** a vector in both has third entry 0 and first and second entries 0, so $\mathcal{S}_1 \cap \mathcal{S}_2 = \{\underset{\sim}{0}\}$.

So $\mathbb{R}^3 = \mathcal{S}_1 \oplus \mathcal{S}_2$.

Example 0.8 (Two planes that sum to $\mathbb{R}^3$ but not directly)**.** The planes $z = 0$ and $x = 0$ sum to $\mathbb{R}^3$ (Example 0.3), but they share the y-axis, so they are not complementary, and $\mathbb{R}^3$ is not their direct sum. The shared line shows up as more than one way to split a vector:

$$(0,1,0) = (0,1,0) + \underset{\sim}{0}$$
$$= \underset{\sim}{0} + (0,1,0),$$

with the first term in the plane $z = 0$ and the second in the plane $x = 0$ both times.

Exercise 0.6 (Testing whether two lines are complementary in $\mathbb{R}^2$)**.** Two subspaces are complementary in $\mathcal{V}$ when their sum is $\mathcal{V}$ and they share only $\underset{\sim}{0}$. Let $\mathcal{S}_1 = \text{span}\{(1,0)\}$ and $\mathcal{S}_2 = \text{span}\{(1,1)\}$.
Decide whether $\mathcal{S}_1$ and $\mathcal{S}_2$ are complementary in $\mathbb{R}^2$.

Hint

Solve $a\,(1,0) = b\,(1,1)$, and express (x,y) as $a\,(1,0) + b\,(1,1)$.

Solution

Solution 0.6. If $a\,(1,0) = b\,(1,1)$, then $(a,0) = (b,b)$, so $b = 0$ and then $a = 0$. So $\mathcal{S}_1 \cap \mathcal{S}_2 = \{\underset{\sim}{0}\}$. Any $(x,y) \in \mathbb{R}^2$ is $(x-y)\,(1,0) + y\,(1,1)$, so $\mathcal{S}_1 + \mathcal{S}_2 = \mathbb{R}^2$. Both conditions hold, so $\mathcal{S}_1$ and $\mathcal{S}_2$ are complementary in $\mathbb{R}^2$.

Exercise 0.7 (Testing whether $\mathbb{R}^3$ is the direct sum of a plane and a tilted line)**.** The direct sum $\mathcal{S}_1 \oplus \mathcal{S}_2$ needs the two subspaces to be complementary. Let $\mathcal{S}_1 = \{(a,b,0) : a,b \in \mathbb{R}\}$, the plane $z = 0$, and $\mathcal{S}_2 = \text{span}\{(1,1,1)\}$, a tilted line.
Decide whether $\mathbb{R}^3 = \mathcal{S}_1 \oplus \mathcal{S}_2$.

Hint

To write (x,y,z) as a vector of $\mathcal{S}_1$ plus $c\,(1,1,1)$, the third entry fixes c.

Solution

Solution 0.7. Any (x,y,z) is $(x-z, y-z, 0) + z\,(1,1,1)$, with the first term in $\mathcal{S}_1$ and the second in $\mathcal{S}_2$, so $\mathcal{S}_1 + \mathcal{S}_2 = \mathbb{R}^3$. Also, $c\,(1,1,1)$ is in $\mathcal{S}_1$ only if its third entry c is 0, so $\mathcal{S}_1 \cap \mathcal{S}_2 = \{\underset{\sim}{0}\}$. So $\mathcal{S}_1$ and $\mathcal{S}_2$ are complementary in $\mathbb{R}^3$, and $\mathbb{R}^3 = \mathcal{S}_1 \oplus \mathcal{S}_2$.

Theorem 0.4 (Three ways to recognize a direct sum)**.** *Let $\mathcal{S}_1$ and $\mathcal{S}_2$ be subspaces of $\mathbb{R}^p$, and let $\mathcal{V} = \mathcal{S}_1 + \mathcal{S}_2$. The following statements are equivalent:*

1. $\mathcal{V} = \mathcal{S}_1 \oplus \mathcal{S}_2$ *(Definition 0.3).*
2. $\dim(\mathcal{V}) = \dim(\mathcal{S}_1) + \dim(\mathcal{S}_2)$.
3. *Every $\underset{\sim}{x} \in \mathcal{V}$ can be written as $\underset{\sim}{x} = \underset{\sim}{x}_1 + \underset{\sim}{x}_2$ with $\underset{\sim}{x}_1 \in \mathcal{S}_1$ and $\underset{\sim}{x}_2 \in \mathcal{S}_2$ in only one way.*

Proof

Proof. Since $\mathcal{V} = \mathcal{S}_1 + \mathcal{S}_2$ is given, statement 1 says exactly that $\mathcal{S}_1 \cap \mathcal{S}_2 = \{\underset{\sim}{0}\}$.
1 and 2 are equivalent. By Corollary 0.1, $\dim(\mathcal{S}_1 + \mathcal{S}_2) = \dim(\mathcal{S}_1) + \dim(\mathcal{S}_2)$ exactly when $\mathcal{S}_1 \cap \mathcal{S}_2 = \{\underset{\sim}{0}\}$.
1 implies 3. Every $\underset{\sim}{x} \in \mathcal{V}$ has at least one such expression, because $\mathcal{V} = \mathcal{S}_1 + \mathcal{S}_2$. Suppose

$$\begin{aligned} \underset{\sim}{x} &= \underset{\sim}{u}_1 + \underset{\sim}{u}_2 \\ &= \underset{\sim}{v}_1 + \underset{\sim}{v}_2, \end{aligned}$$

with $\underset{\sim}{u}_1, \underset{\sim}{v}_1 \in \mathcal{S}_1$ and $\underset{\sim}{u}_2, \underset{\sim}{v}_2 \in \mathcal{S}_2$. Then

$$\begin{aligned} \underset{\sim}{u}_1 - \underset{\sim}{v}_1 &= (\underset{\sim}{u}_1 + \underset{\sim}{u}_2) - \underset{\sim}{u}_2 - \underset{\sim}{v}_1 && \text{(add and subtract } \underset{\sim}{u}_2) \\ &= (\underset{\sim}{v}_1 + \underset{\sim}{v}_2) - \underset{\sim}{u}_2 - \underset{\sim}{v}_1 && \text{(substitute } \underset{\sim}{u}_1 + \underset{\sim}{u}_2 = \underset{\sim}{v}_1 + \underset{\sim}{v}_2) \\ &= \underset{\sim}{v}_2 - \underset{\sim}{u}_2. && \text{(cancel } \underset{\sim}{v}_1) \end{aligned}$$

The left side is in $\mathcal{S}_1$ and the right side is in $\mathcal{S}_2$, because each subspace is closed under addition and scalar multiplication (the difference is $\underset{\sim}{u}_1 + (-1)\,\underset{\sim}{v}_1$). So both sides are in $\mathcal{S}_1 \cap \mathcal{S}_2 = \{\underset{\sim}{0}\}$, which gives $\underset{\sim}{u}_1 = \underset{\sim}{v}_1$ and $\underset{\sim}{u}_2 = \underset{\sim}{v}_2$.
3 implies 1. Take any $\underset{\sim}{x} \in \mathcal{S}_1 \cap \mathcal{S}_2$. Then $\underset{\sim}{x} = \underset{\sim}{x} + \underset{\sim}{0}$, with $\underset{\sim}{x} \in \mathcal{S}_1$ and $\underset{\sim}{0} \in \mathcal{S}_2$, and $\underset{\sim}{x} = \underset{\sim}{0} + \underset{\sim}{x}$, with $\underset{\sim}{0} \in \mathcal{S}_1$ and $\underset{\sim}{x} \in \mathcal{S}_2$ (Theorem 1 in Subspaces and Rank). By statement 3 these two expressions are the same, so $\underset{\sim}{x} = \underset{\sim}{0}$, and $\mathcal{S}_1 \cap \mathcal{S}_2 = \{\underset{\sim}{0}\}$. □

Example 0.9 (Checking the three statements)**.** For the plane $z = 0$ and the z-axis in Example 0.7:

- the dimensions add up:

  $$\begin{aligned} 2 + 1 &= 3 \\ &= \dim(\mathbb{R}^3), \end{aligned}$$

  since the plane has dimension 2 (Example 16 in Subspaces and Rank), the z-axis has the single nonzero vector $(0, 0, 1)$ as a basis, and $\dim(\mathbb{R}^3) = 3$ (Example 16 in Subspaces and Rank);

- the split is unique: if

  $$\begin{aligned} (x, y, z) &= (a, b, 0) + (0, 0, c) \\ &= (a, b, c), \end{aligned}$$

  then $a = x$, $b = y$ and $c = z$.

For the planes $z = 0$ and $x = 0$ in Example 0.8, both statements fail:

$$\begin{aligned} 2 + 2 &= 4 \\ &\neq 3 \\ &= \dim(\mathcal{S}_1 + \mathcal{S}_2) \end{aligned}$$

(Example 0.4), and $(0, 1, 0)$ splits in two ways.

Exercise 0.8 (Recognizing a direct sum by counting dimensions)**.** Statement 2 of Theorem 0.4 recognizes a direct sum from dimensions alone. Let $\mathcal{S}_1 = \text{span}\{(1, 0, 0), (0, 1, 0)\}$ and $\mathcal{S}_2 = \text{span}\{(0, 1, 1)\}$, and suppose that you have already checked that $\mathcal{S}_1 + \mathcal{S}_2 = \mathbb{R}^3$. Decide whether $\mathbb{R}^3 = \mathcal{S}_1 \oplus \mathcal{S}_2$, using statement 2 of Theorem 0.4.

Hint

Find $\dim(\mathcal{S}_1)$ and $\dim(\mathcal{S}_2)$, add them, and compare the total with $\dim(\mathbb{R}^3) = 3$.

Solution

Solution 0.8. Both $(1, 0, 0)$ and $(0, 1, 0)$ lie in $\mathcal{S}_1$, and they are linearly independent, so they are a basis of $\mathcal{S}_1$ and $\dim(\mathcal{S}_1) = 2$. The single nonzero vector $(0, 1, 1)$ is a basis of $\mathcal{S}_2$, so $\dim(\mathcal{S}_2) = 1$. Then

$$\dim(\mathcal{S}_1) + \dim(\mathcal{S}_2) = 2 + 1 = 3 = \dim(\mathbb{R}^3),$$

so by statement 2 of Theorem 0.4, $\mathbb{R}^3 = \mathcal{S}_1 \oplus \mathcal{S}_2$.

Definition 0.4 (Projection onto a subspace along a complement)**.** Let $\mathcal{S}$ and $\mathcal{T}$ be subspaces of $\mathbb{R}^p$ with $\mathbb{R}^p = \mathcal{S} \oplus \mathcal{T}$ (Definition 0.3), and let $y \in \mathbb{R}^p$. By Theorem 0.4, $y = u + v$ for exactly one $u \in \mathcal{S}$ and $v \in \mathcal{T}$. The vector u is the **projection** [3] of y onto $\mathcal{S}$ along $\mathcal{T}$ [4], Definition 6.3, p. 163.

Example 0.10 (The projection depends on the complement)**.** Let $\mathcal{S} = \text{span}\{(1, 0)\}$, the horizontal axis in $\mathbb{R}^2$, and let $y = (3, 2)$.

- **Along** $\mathcal{T}_1 = \text{span}\{(1, -1)\}$**:** $(3, 2) = (5, 0) + (-2, 2)$, with $(5, 0) \in \mathcal{S}$ and $(-2, 2) = -2 \cdot (1, -1) \in \mathcal{T}_1$, so the projection of $(3, 2)$ onto $\mathcal{S}$ along $\mathcal{T}_1$ is $(5, 0)$.
- **Along** $\mathcal{T}_2 = \text{span}\{(0, 1)\}$**:** $(3, 2) = (3, 0) + (0, 2)$, with $(3, 0) \in \mathcal{S}$ and $(0, 2) \in \mathcal{T}_2$, so the projection of $(3, 2)$ onto $\mathcal{S}$ along $\mathcal{T}_2$ is $(3, 0)$.

Each of $\mathcal{T}_1$ and $\mathcal{T}_2$ is a line that meets $\mathcal{S}$ only at 0, so

$$\begin{aligned}\mathbb{R}^2 &= \mathcal{S} \oplus \mathcal{T}_1 \\ &= \mathcal{S} \oplus \mathcal{T}_2\end{aligned}$$

(Definition 0.3, Theorem 0.4). The same vector has a different projection onto the same subspace when the complement changes.

Exercise 0.9 (Projecting onto an axis along a diagonal line)**.** A projection onto $\mathcal{S}$ along $\mathcal{T}$ keeps the piece of a vector that lies in $\mathcal{S}$ when the vector is split along a direct sum. Let $\mathcal{S} = \text{span}\{(1,0)\}$, the horizontal axis in $\mathbb{R}^2$, let $\mathcal{T} = \text{span}\{(1,1)\}$, a diagonal line, and let $y = (4,3)$.
Find the projection of $\underset{\sim}{y}$ onto $\mathcal{S}$ along $\mathcal{T}$.

Hint

Split $(4,3)$ as $(a,0) + c\,(1,1)$, and use the second entry first.

Solution

Solution 0.9. Split $(4,3)$ as $(a,0) + c\,(1,1) = (a+c, c)$. The second entry gives $c = 3$, and then the first gives $a = 1$. So $(4,3) = (1,0) + (3,3)$, with $(1,0) \in \mathcal{S}$ and $(3,3) \in \mathcal{T}$, and the projection of $\underset{\sim}{y}$ onto $\mathcal{S}$ along $\mathcal{T}$ is $(1,0)$.

Orthogonal complements

Theorem 0.1 (The dot product is linear in each slot)**.** *For vectors* $\underset{\sim}{x}, \underset{\sim}{u}, \underset{\sim}{w} \in \mathbb{R}^p$ *and numbers* a, b*,*

$$\underset{\sim}{x} \cdot (a\,\underset{\sim}{u} + b\,\underset{\sim}{w}) = a\,(\underset{\sim}{x} \cdot \underset{\sim}{u}) + b\,(\underset{\sim}{x} \cdot \underset{\sim}{w})$$

and

$$(a\,\underset{\sim}{u} + b\,\underset{\sim}{w}) \cdot \underset{\sim}{x} = a\,(\underset{\sim}{u} \cdot \underset{\sim}{x}) + b\,(\underset{\sim}{w} \cdot \underset{\sim}{x}).$$

Proof

Proof. **Second slot.**

$$
\begin{aligned}
\underset{\sim}{x} \cdot (a\,\underset{\sim}{u} + b\,\underset{\sim}{w}) &= \sum_{i=1}^{p} x_i\,(a\,\underset{\sim}{u} + b\,\underset{\sim}{w})_i && \text{(Definition 8 in Vectors)} \\
&= \sum_{i=1}^{p} x_i\,(a u_i + b w_i) && \text{(Definition 9 in Matrices, Definition 7 in Vectors)} \\
&= \sum_{i=1}^{p} (x_i a u_i + x_i b w_i) && \text{(distribute each } x_i\text{)} \\
&= \sum_{i=1}^{p} (a\,x_i u_i + b\,x_i w_i) && \text{(commute the factors in each product)} \\
&= \sum_{i=1}^{p} a\,x_i u_i + \sum_{i=1}^{p} b\,x_i w_i && \text{(split the finite sum)} \\
&= a \sum_{i=1}^{p} x_i u_i + b \sum_{i=1}^{p} x_i w_i && \text{(factor } a \text{ and } b \text{ out of the sums)} \\
&= a\,(\underset{\sim}{x} \cdot \underset{\sim}{u}) + b\,(\underset{\sim}{x} \cdot \underset{\sim}{w}). && \text{(Definition 8 in Vectors)}
\end{aligned}
$$

First slot.

$$
\begin{aligned}
(a\,\underset{\sim}{u} + b\,\underset{\sim}{w}) \cdot \underset{\sim}{x} &= \underset{\sim}{x} \cdot (a\,\underset{\sim}{u} + b\,\underset{\sim}{w}) && \text{(Theorem 1 in Vectors)} \\
&= a\,(\underset{\sim}{x} \cdot \underset{\sim}{u}) + b\,(\underset{\sim}{x} \cdot \underset{\sim}{w}) && \text{(second slot)} \\
&= a\,(\underset{\sim}{u} \cdot \underset{\sim}{x}) + b\,(\underset{\sim}{w} \cdot \underset{\sim}{x}). && \text{(Theorem 1 in Vectors, twice)}
\end{aligned}
$$

Two special cases are used often. With

$$
\begin{aligned}
a &= b \\
&= 1,
\end{aligned}
$$

and $1\,\underset{\sim}{u} = \underset{\sim}{u}$, the theorem gives $\underset{\sim}{x} \cdot (\underset{\sim}{u} + \underset{\sim}{w}) = \underset{\sim}{x} \cdot \underset{\sim}{u} + \underset{\sim}{x} \cdot \underset{\sim}{w}$. With $\underset{\sim}{w} = \underset{\sim}{u}$ and $b = 0$, and $0\,\underset{\sim}{u} = \underset{\sim}{0}$, it gives $\underset{\sim}{x} \cdot (a\,\underset{\sim}{u}) = a\,(\underset{\sim}{x} \cdot \underset{\sim}{u})$, and likewise in the first slot. □

Example 0.1 (Splitting a dot product)**.** Let $\underset{\sim}{x} = (1, 2)$, $\underset{\sim}{u} = (3, 0)$, $\underset{\sim}{w} = (0, 1)$, $a = 2$ and $b = -1$. Directly, $a\,\underset{\sim}{u} + b\,\underset{\sim}{w} = (6, -1)$ and

$$
\begin{aligned}
\underset{\sim}{x} \cdot (6, -1) &= 6 - 2 \\
&= 4.
\end{aligned}
$$

By Theorem 0.1,

$$
\begin{aligned}
a\,(\underset{\sim}{x} \cdot \underset{\sim}{u}) + b\,(\underset{\sim}{x} \cdot \underset{\sim}{w}) &= 2 \cdot 3 + (-1) \cdot 2 \\
&= 4,
\end{aligned}
$$

the same number.

Exercise 0.1 (Finding a dot product from two known dot products)**.** Theorem 0.1 lets a dot product with a combination be found from the dot products with its parts. Suppose that $\underset{\sim}{x} \cdot \underset{\sim}{u} = 3$ and $\underset{\sim}{x} \cdot \underset{\sim}{w} = -1$.
Find $\underset{\sim}{x} \cdot (2\,\underset{\sim}{u} + 5\,\underset{\sim}{w})$.

Hint

Use Theorem 0.1 with $a = 2$ and $b = 5$.

Solution

Solution 0.1. By Theorem 0.1,

$$\underset{\sim}{x} \cdot (2\,\underset{\sim}{u} + 5\,\underset{\sim}{w}) = 2\,(\underset{\sim}{x} \cdot \underset{\sim}{u}) + 5\,(\underset{\sim}{x} \cdot \underset{\sim}{w}) = 2 \cdot 3 + 5 \cdot (-1) = 1.$$

Definition 0.1 (Orthogonal complement)**.** The **orthogonal complement** [5] of a set $\mathcal{X}$ of vectors in $\mathbb{R}^p$ is the set of vectors orthogonal (Definition 20 in Vectors) to every vector of $\mathcal{X}$:

$$\mathcal{X}^{\perp} \stackrel{\text{def}}{=} \{\underset{\sim}{u} \in \mathbb{R}^p : \underset{\sim}{x} \cdot \underset{\sim}{u} = 0 \text{ for all } \underset{\sim}{x} \in \mathcal{X}\}.$$

$\mathcal{X}^{\perp}$ is read "$\mathcal{X}$ perp".

Example 0.2 (Orthogonal complements in $\mathbb{R}^3$)**.**

- **A plane.** Let $\mathcal{X} = \{(a, b, 0) : a, b \in \mathbb{R}\}$, the plane $z = 0$. A vector $\underset{\sim}{u}$ is in $\mathcal{X}^{\perp}$ when

$$\begin{aligned}(a, b, 0) \cdot \underset{\sim}{u} &= au_1 + bu_2 \\ &= 0\end{aligned}$$

  for all a and b. Taking $(a, b) = (1, 0)$ forces $u_1 = 0$, and taking $(a, b) = (0, 1)$ forces $u_2 = 0$; conversely, if

$$\begin{aligned}u_1 &= u_2 \\ &= 0,\end{aligned}$$

  then $au_1 + bu_2 = 0$ for all a and b. So $\mathcal{X}^{\perp} = \{(0, 0, c) : c \in \mathbb{R}\}$, the z-axis.

- **Two vectors.** The two-element set $\{(1, 0, 0), (0, 1, 0)\}$ is not a subspace, and its orthogonal complement is the z-axis too: the two conditions $u_1 = 0$ and $u_2 = 0$ are the same as for the plane.

- **The extremes.** $\{\underset{\sim}{0}\}^\perp = \mathbb{R}^3$, because $\underset{\sim}{0} \cdot \underset{\sim}{u} = 0$ for every $\underset{\sim}{u}$. $(\mathbb{R}^3)^\perp = \{\underset{\sim}{0}\}$, because a $\underset{\sim}{u}$ in it is orthogonal to itself, and

$$\begin{aligned} \underset{\sim}{u} \cdot \underset{\sim}{u} &= u_1^2 + u_2^2 + u_3^2 \\ &= 0 \end{aligned}$$

forces $\underset{\sim}{u} = \underset{\sim}{0}$.

Exercise 0.2 (Testing whether a vector is in an orthogonal complement)**.** A vector is in the orthogonal complement of a set when it is orthogonal to every vector of the set. Let $\mathcal{X} = \{(1, 2, 0), (0, 3, 1)\}$, a set of two vectors in $\mathbb{R}^3$.
Decide whether the vector $(2, -1, 3)$ is in $\mathcal{X}^\perp$.

Hint

Take the dot product of $(2, -1, 3)$ with each vector of $\mathcal{X}$.

Solution

Solution 0.2. Here

$$\begin{aligned} (1, 2, 0) \cdot (2, -1, 3) &= 2 - 2 + 0 = 0, \\ (0, 3, 1) \cdot (2, -1, 3) &= 0 - 3 + 3 = 0, \end{aligned}$$

so $(2, -1, 3)$ is orthogonal to both vectors of $\mathcal{X}$, and it is in $\mathcal{X}^\perp$.

Remark (The orthogonal complement of a set that is not a subspace)

Remark 0.1 (The orthogonal complement of a set that is not a subspace). $\mathcal{X}$ in Definition 0.1 need not be a subspace.

Theorem 0.2 (An orthogonal complement is a subspace)**.** *For any set $\mathcal{X}$ of vectors in $\mathbb{R}^p$, $\mathcal{X}^\perp$ (Definition 0.1) is a subspace of $\mathbb{R}^p$ (Definition 4 in Subspaces and Rank).*

Proof

Proof. $\mathcal{X}^\perp$ is not empty:

$$\begin{aligned} \underset{\sim}{x} \cdot \underset{\sim}{0} &= \sum_i x_i \cdot 0 \\ &= 0 \end{aligned}$$

for every $\underset{\sim}{x}$, so $\underset{\sim}{0} \in \mathcal{X}^\perp$. Let $\underset{\sim}{u}, \underset{\sim}{w} \in \mathcal{X}^\perp$, let $c \in \mathbb{R}$, and let $\underset{\sim}{x} \in \mathcal{X}$.
Addition.

$$\begin{aligned} \underset{\sim}{x} \cdot (\underset{\sim}{u} + \underset{\sim}{w}) &= \underset{\sim}{x} \cdot \underset{\sim}{u} + \underset{\sim}{x} \cdot \underset{\sim}{w} && \text{(Theorem 0.1, special case } a \text{ and } b \text{ both equal to 1)} \\ &= 0 + 0 && (\underset{\sim}{u}, \underset{\sim}{w} \in \mathcal{X}^\perp) \\ &= 0. && \text{(arithmetic)} \end{aligned}$$

Scalar multiplication.

$$\begin{aligned} \underset{\sim}{x} \cdot (c\,\underset{\sim}{u}) &= c\,(\underset{\sim}{x} \cdot \underset{\sim}{u}) && \text{(Theorem 0.1, special case } b = 0) \\ &= c \cdot 0 && (\underset{\sim}{u} \in \mathcal{X}^\perp) \\ &= 0. && \text{(arithmetic)} \end{aligned}$$

These two equations hold for every $\underset{\sim}{x} \in \mathcal{X}$, so $\underset{\sim}{u} + \underset{\sim}{w}$ and $c\,\underset{\sim}{u}$ are in $\mathcal{X}^\perp$. □

Example 0.3 (The complement of a single vector is a line)**.** The one-element set $\mathcal{X} = \{(1, 2)\}$ in $\mathbb{R}^2$ is not a subspace: $2\,(1, 2) = (2, 4)$ is not in it. Its orthogonal complement is

$$\begin{aligned} \mathcal{X}^\perp &= \{\underset{\sim}{u} \in \mathbb{R}^2 : u_1 + 2u_2 = 0\} \\ &= \{c\,(-2, 1) : c \in \mathbb{R}\}, \end{aligned}$$

since $u_1 + 2u_2 = 0$ means

$$\begin{aligned} \underset{\sim}{u} &= (-2u_2, u_2) \\ &= u_2\,(-2, 1). \end{aligned}$$

That set is span$\{(-2, 1)\}$, a subspace by Theorem 2 in Subspaces and Rank, as Theorem 0.2 says it must be. It contains $(-2, 1)$, the vector Remark 8 in Vectors found perpendicular to $(1, 2)$.

Exercise 0.3 (Finding the orthogonal complement of two vectors)**.** Theorem 0.2 says that an orthogonal complement is always a subspace, even when the original set is not. Let $\mathcal{X} = \{(1, 0, 1), (1, 0, -1)\}$, a set of two vectors in $\mathbb{R}^3$.
Find $\mathcal{X}^\perp$ as a span.

Hint

A vector $\underset{\sim}{u}$ is in $\mathcal{X}^\perp$ when $u_1 + u_3 = 0$ and $u_1 - u_3 = 0$.

Solution

Solution 0.3. A vector $\underset{\sim}{u}$ is in $\mathcal{X}^\perp$ exactly when $(1,0,1)\cdot\underset{\sim}{u} = u_1+u_3 = 0$ and $(1,0,-1)\cdot\underset{\sim}{u} = u_1 - u_3 = 0$. Adding the two equations gives $u_1 = 0$, and then $u_3 = 0$, while u_2 is free. So

$$\mathcal{X}^\perp = \{(0,u_2,0) : u_2 \in \mathbb{R}\} = \text{span}\{(0,1,0)\},$$

the second coordinate axis, which is a subspace, as Theorem 0.2 says.

Theorem 0.3 (The orthogonal complement of a column space is a null space)**.** *For any* $m \times n$ *matrix* $\mathbf{A}$*,*

$$\mathcal{C}(\mathbf{A})^\perp = \mathcal{N}(\mathbf{A}^\top).$$

Proof

Proof. Let $\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_n$ be the columns of $\mathbf{A}$, so $\underset{\sim}{a}_j = (a_{1j}, \ldots, a_{mj})$. For $\underset{\sim}{x} \in \mathbb{R}^m$, entry j of $\mathbf{A}^\top\underset{\sim}{x}$ is

$$\begin{aligned}(\mathbf{A}^\top\underset{\sim}{x})_j &= \sum_{i=1}^{m}(\mathbf{A}^\top)_{ji}\,x_i && \text{(Definition 16 in Matrices)}\\ &= \sum_{i=1}^{m} a_{ij}\,x_i && \text{(Definition 6 in Matrices)}\\ &= \underset{\sim}{a}_j \cdot \underset{\sim}{x}. && \text{(Definition 8 in Vectors)}\end{aligned}$$

So $\underset{\sim}{x} \in \mathcal{N}(\mathbf{A}^\top)$ (Definition 3 in Column Space, Null Space and Rank-Nullity) exactly when $\underset{\sim}{a}_j \cdot \underset{\sim}{x} = 0$ for every column $\underset{\sim}{a}_j$.
$\mathcal{C}(\mathbf{A})^\perp \subseteq \mathcal{N}(\mathbf{A}^\top)$**.** Each column $\underset{\sim}{a}_j$ is in $\mathcal{C}(\mathbf{A})$ (Theorem 1 in Column Space, Null Space and Rank-Nullity), so a vector orthogonal to all of $\mathcal{C}(\mathbf{A})$ is orthogonal to every column.
$\mathcal{N}(\mathbf{A}^\top) \subseteq \mathcal{C}(\mathbf{A})^\perp$**.** Suppose $\underset{\sim}{a}_j \cdot \underset{\sim}{x} = 0$ for every j, and take any $\underset{\sim}{y} \in \mathcal{C}(\mathbf{A})$, so $\underset{\sim}{y} = \mathbf{A}\underset{\sim}{z}$ for some $\underset{\sim}{z} \in \mathbb{R}^n$ (Definition 1 in Column Space, Null Space and Rank-Nullity). Then

$$
\begin{aligned}
\underset{\sim}{y} \cdot \underset{\sim}{x} &= \sum_{i=1}^{m} (\mathbf{A}\underset{\sim}{z})_i \, x_i && \text{(Definition 8 in Vectors)} \\
&= \sum_{i=1}^{m} \left(\sum_{j=1}^{n} a_{ij} z_j \right) x_i && \text{(Definition 16 in Matrices)} \\
&= \sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij} \, z_j \, x_i && \text{(distribute each } x_i \text{ over the inner sum)} \\
&= \sum_{j=1}^{n} \sum_{i=1}^{m} a_{ij} \, z_j \, x_i && \text{(swap the order of the finite sums)} \\
&= \sum_{j=1}^{n} z_j \sum_{i=1}^{m} a_{ij} \, x_i && \text{(commute, then factor } z_j \text{ out of the inner sum)} \\
&= \sum_{j=1}^{n} z_j \, (\underset{\sim}{a}_j \cdot \underset{\sim}{x}) && \text{(Definition 8 in Vectors)} \\
&= \sum_{j=1}^{n} z_j \cdot 0 && \text{(each } \underset{\sim}{a}_j \cdot \underset{\sim}{x} = 0\text{)} \\
&= 0. && \text{(arithmetic)}
\end{aligned}
$$

So $\underset{\sim}{x}$ is orthogonal to every vector of $\mathcal{C}(\mathbf{A})$. □

Example 0.4 (The complement of the column space in Example 1 in Column Space, Null Space and Rank-Nullity)**.** For $\mathbf{A}$ in Example 1 in Column Space, Null Space and Rank-Nullity, $\mathcal{C}(\mathbf{A}) = \text{span}\{(1, 3)\}$ (1). Example 3 in Column Space, Null Space and Rank-Nullity found $\mathcal{N}(\mathbf{A}^\top) = \text{span}\{(-3, 1)\}$, so by Theorem 0.3 this line is $\mathcal{C}(\mathbf{A})^\perp$. Directly, for any c,

$$
\begin{aligned}
(c\,(1,3)) \cdot (-3,1) &= c\,((1,3) \cdot (-3,1)) && \text{(Theorem 0.1, special case } b = 0\text{, first slot)} \\
&= c\,(-3+3) && \text{(Definition 8 in Vectors)} \\
&= 0. && \text{(arithmetic)}
\end{aligned}
$$

Exercise 0.4 (Finding the orthogonal complement of a column space)**.** Theorem 0.3 turns the orthogonal complement of a column space into a null space, which a linear system describes. Let

$$
\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}.
$$

Find $\mathcal{C}(\mathbf{A})^\perp$ as a span.

Hint

Solve $\mathbf{A}^\top \underset{\sim}{y} = \underset{\sim}{0}_2$.

Solution

Solution 0.4. By Theorem 0.3, $\mathcal{C}(\mathbf{A})^\perp = \mathcal{N}(\mathbf{A}^\top)$. For $\underset{\sim}{y} = (y_1, y_2, y_3)$,

$$\mathbf{A}^\top \underset{\sim}{y} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} y_1 + y_3 \\ y_2 + y_3 \end{bmatrix},$$

which is $\underset{\sim}{0}_2$ exactly when $y_1 = y_2 = -y_3$. So

$$\mathcal{C}(\mathbf{A})^\perp = \text{span}\{(-1, -1, 1)\}.$$

As a check, $(-1, -1, 1)$ is orthogonal to both columns of $\mathbf{A}$: $-1+0+1 = 0$ and $0-1+1 = 0$.

Theorem 0.4 (A subspace and its orthogonal complement make up the whole space)**.** *Let $\mathcal{S}$ be a subspace of $\mathbb{R}^p$. Then*

1. $\mathcal{S} \cap \mathcal{S}^\perp = \{\underset{\sim}{0}\}$;
2. $\dim(\mathcal{S}) + \dim(\mathcal{S}^\perp) = p$;
3. $\mathbb{R}^p = \mathcal{S} \oplus \mathcal{S}^\perp$ *(Definition 0.3).*

So every $\underset{\sim}{y} \in \mathbb{R}^p$ can be written as $\underset{\sim}{y} = \underset{\sim}{u} + \underset{\sim}{v}$ with $\underset{\sim}{u} \in \mathcal{S}$ and $\underset{\sim}{v} \in \mathcal{S}^\perp$ in only one way (Theorem 0.4).

Proof

Proof. $\mathcal{S}^\perp$ is a subspace (Theorem 0.2), so it has a dimension (Definition 9 in Subspaces and Rank).

Part 1. Both $\mathcal{S}$ and $\mathcal{S}^\perp$ contain $\underset{\sim}{0}$ (Theorem 1 in Subspaces and Rank). If $\underset{\sim}{x}$ is in both, then $\underset{\sim}{x}$ is orthogonal to every vector of $\mathcal{S}$, $\underset{\sim}{x}$ itself included, so

$$\begin{aligned} \underset{\sim}{x} \cdot \underset{\sim}{x} &= x_1^2 + \cdots + x_p^2 \\ &= 0, \end{aligned}$$

which forces every $x_i = 0$.

Part 2. Let $d = \dim(\mathcal{S})$. If $d = 0$, then $\mathcal{S}$ has a basis (Theorem 9 in Subspaces and Rank) with 0 vectors (Definition 9 in Subspaces and Rank), the empty list, whose span is $\{\underset{\sim}{0}\}$ (Definition 5 in Subspaces and Rank); so $\mathcal{S} = \{\underset{\sim}{0}\}$, $\mathcal{S}^\perp = \mathbb{R}^p$ (every $\underset{\sim}{u}$ satisfies $\underset{\sim}{0} \cdot \underset{\sim}{u} = 0$), and $0 + p = p$ (Example 16 in Subspaces and Rank). If $d \geq 1$, choose a

basis $\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_d$ of $\mathcal{S}$ (Theorem 9 in Subspaces and Rank), and let $\mathbf{A}$ be the $p \times d$ matrix with these columns. The basis spans $\mathcal{S}$, so $\mathcal{S} = \mathcal{C}(\mathbf{A})$ (Theorem 1 in Column Space, Null Space and Rank-Nullity), and $\mathcal{S}^\perp = \mathcal{N}(\mathbf{A}^\top)$ (Theorem 0.3). $\mathbf{A}^\top$ is $d \times p$, so

$$\begin{aligned}
\dim(\mathcal{S}^\perp) &= \dim(\mathcal{N}(\mathbf{A}^\top)) && \text{(Theorem 0.3)} \\
&= \text{nullity}(\mathbf{A}^\top) && \text{(Definition 7 in Column Space, Null Space and Rank-Nullity)} \\
&= p - \text{rank}(\mathbf{A}^\top) && \text{(Theorem 8 in Column Space, Null Space and Rank-Nullity, for the } p \text{ columns o} \\
&= p - \text{rank}(\mathbf{A}) && \text{(Theorem 10 in Column Space, Null Space and Rank-Nullity)} \\
&= p - \dim(\mathcal{C}(\mathbf{A})) && \text{(Theorem 7 in Column Space, Null Space and Rank-Nullity)} \\
&= p - d. && (\mathcal{C}(\mathbf{A}) = \mathcal{S})
\end{aligned}$$

Part 3. $\mathcal{S} + \mathcal{S}^\perp$ is a subspace of $\mathbb{R}^p$ (Theorem 0.1), and

$$\begin{aligned}
\dim(\mathcal{S} + \mathcal{S}^\perp) &= \dim(\mathcal{S}) + \dim(\mathcal{S}^\perp) - \dim(\mathcal{S} \cap \mathcal{S}^\perp) && \text{(Theorem 0.2)} \\
&= p - \dim(\mathcal{S} \cap \mathcal{S}^\perp) && \text{(part 2)} \\
&= p - 0 && \text{(part 1, and } \dim(\{\underset{\sim}{0}\}) = 0 \text{ by Example 16 in Subspaces a} \\
&= p. && \text{(arithmetic)}
\end{aligned}$$

Since $\mathcal{S} + \mathcal{S}^\perp \subseteq \mathbb{R}^p$, $\mathbb{R}^p$ is a subspace (Example 3 in Subspaces and Rank) and $\dim(\mathbb{R}^p) = p$ (Example 16 in Subspaces and Rank), Theorem 11 in Subspaces and Rank gives $\mathcal{S} + \mathcal{S}^\perp = \mathbb{R}^p$. Together with part 1, $\mathbb{R}^p = \mathcal{S} \oplus \mathcal{S}^\perp$. The uniqueness of the split then follows from Theorem 0.4 (statement 1 implies statement 3). □

Example 0.5 (Splitting a vector of $\mathbb{R}^3$ along a line and its complement)**.** Let $\mathcal{S} = \text{span}\{(1,1,0)\}$ in $\mathbb{R}^3$. A single nonzero vector is linearly independent ($c\,\underset{\sim}{v} = \underset{\sim}{0}$ with $\underset{\sim}{v} \neq \underset{\sim}{0}$ forces $c = 0$), so $(1,1,0)$ is a basis of $\mathcal{S}$ and $\dim(\mathcal{S}) = 1$. A vector $\underset{\sim}{u}$ is in $\mathcal{S}^\perp$ exactly when $u_1 + u_2 = 0$: that condition says $(1,1,0) \cdot \underset{\sim}{u} = 0$, and then

$$\begin{aligned}
c\,(1,1,0) \cdot \underset{\sim}{u} &= c\,(u_1 + u_2) \\
&= 0
\end{aligned}$$

for every c. So

$$\begin{aligned}
\mathcal{S}^\perp &= \{(a, -a, c) : a, c \in \mathbb{R}\} \\
&= \text{span}\{(1,-1,0), (0,0,1)\}.
\end{aligned}$$

Those two vectors are linearly independent, because $a\,(1,-1,0) + c\,(0,0,1) = (a,-a,c)$ is $\underset{\sim}{0}$ only if

$$\begin{aligned} a &= c \\ &= 0, \end{aligned}$$

so $\dim(\mathcal{S}^\perp) = 2$, and $1 + 2 = 3$, as part 2 says.
To split $\underset{\sim}{y} = (3, 1, 2)$, look for $\underset{\sim}{u} = t\,(1, 1, 0)$ with

$$\begin{aligned} \underset{\sim}{v} &= \underset{\sim}{y} - \underset{\sim}{u} \\ &= (3 - t, 1 - t, 2) \end{aligned}$$

in $\mathcal{S}^\perp$: that membership needs $(3 - t) + (1 - t) = 0$, so $t = 2$. Then $\underset{\sim}{u} = (2, 2, 0)$ and $\underset{\sim}{v} = (1, -1, 2)$, with $\underset{\sim}{u} + \underset{\sim}{v} = (3, 1, 2)$ and

$$\begin{aligned} (1, 1, 0) \cdot (1, -1, 2) &= 1 - 1 + 0 \\ &= 0. \end{aligned}$$

Exercise 0.5 (Splitting a vector into a piece in a line and an orthogonal piece)**.** By Theorem 0.4, every vector of $\mathbb{R}^p$ splits in only one way into a piece in a subspace $\mathcal{S}$ and a piece in its orthogonal complement. Let $\mathcal{S} = \text{span}\{(1, 2)\}$, a line in $\mathbb{R}^2$, and let $\underset{\sim}{y} = (5, 0)$. Find $\underset{\sim}{u} \in \mathcal{S}$ and $\underset{\sim}{v} \in \mathcal{S}^\perp$ with $\underset{\sim}{y} = \underset{\sim}{u} + \underset{\sim}{v}$.

Hint

Take $\underset{\sim}{u} = t\,(1, 2)$, and choose t so that $\underset{\sim}{y} - \underset{\sim}{u}$ is orthogonal to $(1, 2)$.

Solution

Solution 0.5. Take $\underset{\sim}{u} = t\,(1, 2)$. Then $\underset{\sim}{v} = \underset{\sim}{y} - \underset{\sim}{u} = (5 - t, -2t)$ is in $\mathcal{S}^\perp$ exactly when

$$(1, 2) \cdot (5 - t, -2t) = 5 - t - 4t = 5 - 5t = 0,$$

so $t = 1$. Then $\underset{\sim}{u} = (1, 2)$ and $\underset{\sim}{v} = (4, -2)$, with $\underset{\sim}{u} + \underset{\sim}{v} = (5, 0)$ and $(1, 2) \cdot (4, -2) = 4 - 4 = 0$.

Theorem 0.5 (The orthogonal complement of the orthogonal complement)**.** *For any subspace $\mathcal{S}$ of $\mathbb{R}^p$,*

$$(\mathcal{S}^\perp)^\perp = \mathcal{S}.$$

Proof

Proof. $\mathcal{S} \subseteq (\mathcal{S}^\perp)^\perp$**.** Take $\underset{\sim}{s} \in \mathcal{S}$ and any $\underset{\sim}{v} \in \mathcal{S}^\perp$. By Definition 0.1, $\underset{\sim}{s} \cdot \underset{\sim}{v} = 0$, so $\underset{\sim}{v} \cdot \underset{\sim}{s} = 0$ too (Theorem 1 in Vectors). So $\underset{\sim}{s}$ is orthogonal to every vector of $\mathcal{S}^\perp$, that is, $\underset{\sim}{s} \in (\mathcal{S}^\perp)^\perp$.

Equal dimensions. $\mathcal{S}^\perp$ is a subspace (Theorem 0.2), so part 2 of Theorem 0.4 applies to it as well as to $\mathcal{S}$:

$$\begin{aligned}\dim((\mathcal{S}^\perp)^\perp) &= p - \dim(\mathcal{S}^\perp) && \text{(Theorem 0.4, applied to } \mathcal{S}^\perp\text{)}\\ &= p - (p - \dim(\mathcal{S})) && \text{(Theorem 0.4, applied to } \mathcal{S}\text{)}\\ &= \dim(\mathcal{S}). && \text{(arithmetic)}\end{aligned}$$

$(\mathcal{S}^\perp)^\perp$ is a subspace (Theorem 0.2) that contains $\mathcal{S}$ and has the same dimension, so it equals $\mathcal{S}$ (Theorem 11 in Subspaces and Rank). □

Example 0.6 (Back to the line)**.** In Example 0.5, $\mathcal{S} = \text{span}\{(1,1,0)\}$ and $\mathcal{S}^\perp = \text{span}\{(1,-1,0),(0,0,1)\}$. Put those two spanning vectors in the columns of the 3×2 matrix $\mathbf{B} = \begin{bmatrix} 1 & 0 \\ -1 & 0 \\ 0 & 1 \end{bmatrix}$, so $\mathcal{S}^\perp = \mathcal{C}(\mathbf{B})$ (Theorem 1 in Column Space, Null Space and Rank-Nullity) and $(\mathcal{S}^\perp)^\perp = \mathcal{N}(\mathbf{B}^\top)$ (Theorem 0.3). Since $\mathbf{B}^\top \underset{\sim}{w} = (w_1 - w_2,\ w_3)$ (Definition 16 in Matrices), $\underset{\sim}{w} \in (\mathcal{S}^\perp)^\perp$ exactly when $w_1 - w_2 = 0$ and $w_3 = 0$. So

$$\begin{aligned}(\mathcal{S}^\perp)^\perp &= \{(a,a,0) : a \in \mathbb{R}\}\\ &= \mathcal{S}.\end{aligned}$$

Example 0.7 (A set that is not a subspace does not come back)**.** The one-element set $\mathcal{X} = \{(1,1,0)\}$ is not a subspace. Its orthogonal complement is the same as that of $\mathcal{S}$ in Example 0.6, since both are defined by the single condition $u_1 + u_2 = 0$, so $(\mathcal{X}^\perp)^\perp = \text{span}\{(1,1,0)\}$, which contains $(2,2,0) \notin \mathcal{X}$. Theorem 0.5 needs $\mathcal{S}$ to be a subspace.

Exercise 0.6 (Taking the orthogonal complement twice)**.** Theorem 0.5 says that taking the orthogonal complement of a subspace twice gives back the subspace. Let $\mathcal{S} = \text{span}\{(1,2)\}$ in $\mathbb{R}^2$, whose orthogonal complement is $\mathcal{S}^\perp = \text{span}\{(-2,1)\}$.
Find $(\mathcal{S}^\perp)^\perp$ directly from $\mathcal{S}^\perp = \text{span}\{(-2,1)\}$.

Hint

A vector $\underset{\sim}{w}$ is orthogonal to every multiple of $(-2,1)$ exactly when $-2w_1 + w_2 = 0$.

Solution

Solution 0.6. A vector $\underset{\sim}{w}$ is in $(\mathcal{S}^\perp)^\perp$ exactly when $c\,(-2,1)\cdot\underset{\sim}{w} = c\,(-2w_1+w_2) = 0$ for every c, that is, when $w_2 = 2w_1$. So

$$(\mathcal{S}^\perp)^\perp = \{(w_1, 2w_1) : w_1 \in \mathbb{R}\} = \operatorname{span}\{(1,2)\} = \mathcal{S},$$

as Theorem 0.5 says.

The fundamental theorem of linear algebra

Theorem 0.1 (Fundamental theorem of linear algebra)**.** *Let* $\mathbf{A}$ *be an* $m \times n$ *matrix with* $\operatorname{rank}(\mathbf{A}) = r$*. Then*

1. $\mathcal{C}(\mathbf{A})^\perp = \mathcal{N}(\mathbf{A}^\top)$*, and* $\mathbb{R}^m = \mathcal{C}(\mathbf{A}) \oplus \mathcal{N}(\mathbf{A}^\top)$*;*
2. $\mathcal{C}(\mathbf{A}) = \mathcal{N}(\mathbf{A}^\top)^\perp$*;*
3. $\mathcal{N}(\mathbf{A})^\perp = \mathcal{C}(\mathbf{A}^\top)$*, and* $\mathbb{R}^n = \mathcal{C}(\mathbf{A}^\top) \oplus \mathcal{N}(\mathbf{A})$*;*
4. $$\begin{aligned}\dim(\mathcal{C}(\mathbf{A})) &= \dim(\mathcal{C}(\mathbf{A}^\top)) \\ &= r,\end{aligned}$$

   $\dim(\mathcal{N}(\mathbf{A})) = n - r$*, and* $\dim(\mathcal{N}(\mathbf{A}^\top)) = m - r$*.*

Proof

Proof. **Part 1.** The first equation is Theorem 0.3. $\mathcal{C}(\mathbf{A})$ is a subspace of $\mathbb{R}^m$ (Theorem 1 in Column Space, Null Space and Rank-Nullity), so

$$\begin{aligned}\mathbb{R}^m &= \mathcal{C}(\mathbf{A}) \oplus \mathcal{C}(\mathbf{A})^\perp && \text{(Theorem 0.4)} \\ &= \mathcal{C}(\mathbf{A}) \oplus \mathcal{N}(\mathbf{A}^\top). && \text{(first equation)}\end{aligned}$$

Part 2.

$$\begin{aligned}\mathcal{C}(\mathbf{A}) &= (\mathcal{C}(\mathbf{A})^\perp)^\perp && \text{(Theorem 0.5)} \\ &= \mathcal{N}(\mathbf{A}^\top)^\perp. && \text{(part 1)}\end{aligned}$$

Part 3. Apply parts 2 and 1 to the $n \times m$ matrix $\mathbf{A}^\top$, using $(\mathbf{A}^\top)^\top = \mathbf{A}$: part 2 gives $\mathcal{C}(\mathbf{A}^\top) = \mathcal{N}(\mathbf{A})^\perp$, and part 1 gives $\mathbb{R}^n = \mathcal{C}(\mathbf{A}^\top) \oplus \mathcal{N}(\mathbf{A})$.

Part 4. $\dim(\mathcal{C}(\mathbf{A})) = r$ by Theorem 7 in Column Space, Null Space and Rank-Nullity. $\dim(\mathcal{C}(\mathbf{A}^\top)) = \operatorname{rank}(\mathbf{A}^\top)$ by Theorem 7 in Column Space, Null Space and Rank-Nullity, and that is r by Theorem 10 in Column Space, Null Space and Rank-Nullity. $\mathbf{A}$ has n columns, so $\dim(\mathcal{N}(\mathbf{A})) = n - r$ (Definition 7 in Column Space, Null Space and Rank-Nullity, Theorem 8 in Column Space, Null Space and Rank-Nullity). $\mathbf{A}^\top$ has m columns and rank r, so $\dim(\mathcal{N}(\mathbf{A}^\top)) = m - r$ in the same way. □

Example 0.1 (The four subspaces of the matrix in Example 1 in Column Space, Null Space and Rank-Nullity)**.** For $\mathbf{A}$ in Example 1 in Column Space, Null Space and Rank-Nullity, $m = 2$, $n = 3$ and $r = 1$ (Example 11 in Column Space, Null Space and Rank-Nullity).

- **In $\mathbb{R}^3$:** the row space $\mathcal{C}(\mathbf{A}^\top) = \operatorname{span}\{(1, -2, -2)\}$

  (1) has dimension $1 = r$, and $\mathcal{N}(\mathbf{A}) = \operatorname{span}\{(2, 1, 0), (2, 0, 1)\}$ has dimension $2 = n - r$ (Example 12 in Column Space, Null Space and Rank-Nullity). The spanning vectors are orthogonal:

$$\begin{aligned}(1, -2, -2) \cdot (2, 1, 0) &= 2 - 2 + 0 \\ &= 0\end{aligned}$$

  and

$$\begin{aligned}(1, -2, -2) \cdot (2, 0, 1) &= 2 + 0 - 2 \\ &= 0.\end{aligned}$$

- **In $\mathbb{R}^2$:** $\mathcal{C}(\mathbf{A}) = \operatorname{span}\{(1, 3)\}$ has dimension $1 = r$, and $\mathcal{N}(\mathbf{A}^\top) = \operatorname{span}\{(-3, 1)\}$ has dimension $1 = m - r$ (Example 0.4).

By part 3, $y = (1, 0, 0)$ splits uniquely into a row-space part $t\,(1, -2, -2)$ and a null-space part. The null-space part $y - t\,(1, -2, -2)$ lies in $\mathcal{N}(\mathbf{A})$ (part 3), and $\mathcal{N}(\mathbf{A}) = \mathcal{C}(\mathbf{A}^\top)^\perp$ (Theorem 0.3 applied to $\mathbf{A}^\top$, with $(\mathbf{A}^\top)^\top = \mathbf{A}$), so it must be orthogonal to $(1, -2, -2)$:

$$\begin{aligned}
0 &= (1, -2, -2) \cdot ((1, 0, 0) - t\,(1, -2, -2)) && \text{(orthogonality)} \\
&= (1, -2, -2) \cdot (1 \cdot (1, 0, 0) + (-t)\,(1, -2, -2)) && \text{(write the difference as a linear combination)} \\
&= 1 \cdot ((1, -2, -2) \cdot (1, 0, 0)) + (-t)\,((1, -2, -2) \cdot (1, -2, -2)) && \text{(Theorem 0.1)} \\
&= 1 \cdot 1 + (-t) \cdot 9 && \text{(Definition 8 in Vectors)} \\
&= 1 - 9t, && \text{(arithmetic)}
\end{aligned}$$

so $t = \frac{1}{9}$. The row-space part is $\left(\frac{1}{9}, -\frac{2}{9}, -\frac{2}{9}\right)$ and the null-space part is $\left(\frac{8}{9}, \frac{2}{9}, \frac{2}{9}\right)$; as a check, row 1 of $\mathbf{A}$ gives $\frac{8}{9} - 2 \cdot \frac{2}{9} - 2 \cdot \frac{2}{9} = 0$, and row 2 is 3 times row 1, so $\mathbf{A}$ sends the null-space part to $\underset{\sim}{0}_2$.
These four statements are the fundamental theorem of linear algebra [6].

Exercise 0.1 (Testing whether a linear system has a solution through the left null space)**.** Part 2 of Theorem 0.1 says that $\mathcal{C}(\mathbf{A}) = \mathcal{N}(\mathbf{A}^\top)^\perp$, so a linear system $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$ has a solution exactly when $\underset{\sim}{b}$ is orthogonal to every vector of the left null space. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}, \qquad \underset{\sim}{b} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}.$$

Decide whether $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$ has a solution, by testing $\underset{\sim}{b}$ against $\mathcal{N}(\mathbf{A}^\top)$.

Hint

Find one nonzero vector of $\mathcal{N}(\mathbf{A}^\top)$ whose third entry is 0.

Solution

Solution 0.1. Here $\mathbf{A}^\top \underset{\sim}{y} = (y_1 + y_2, y_1 + y_2)$, so $(1, -1, 0)$ is in $\mathcal{N}(\mathbf{A}^\top)$. But

$$\underset{\sim}{b} \cdot (1, -1, 0) = 1 - 2 + 0 = -1 \neq 0,$$

so $\underset{\sim}{b}$ is not in $\mathcal{N}(\mathbf{A}^\top)^\perp = \mathcal{C}(\mathbf{A})$, and $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$ has no solution. Directly, the first two entries of $\mathbf{A}\underset{\sim}{x}$ are both $x_1 + x_2$, while those of $\underset{\sim}{b}$ are 1 and 2.

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